DSA440/CS540 Group Projects

This is the specification file for the DSA440/CS540 Data and AI Ethics course group projects. Please make sure you read this carefully, especially the deliverables and their deadlines. The group forming sheet is here. You will conduct the group projects with your formed teams. Please mark the “Group Completed” tab with a checkmark if your group has a valid number of people and is final. Only groups with “Group Completed” checkmark will be able to select a domain as specified below.

The general idea of the group project is to find an AI use case that is/might be problematic. For example, AI can be used in mental health help applications. Review the AI use in the selected topic in terms of benefits/harms, potential biases, privacy issues, and other possible issues. Then design a system that will improve AI use in your selected topic in line with ethical AI principles. More details on each step are below.

A. Project Topic Selection

Every team will select an AI use case that is/might be problematic—for example, using AI in loan applications, parole decisions, illness diagnosis, mental health help, and many others. To avoid too many overlaps of the same project topics, please write your selected **domain** in the group forming Excel sheet as soon as possible. Example domains: Health, Finance, Government, Transportation, Hiring, Education, Charity Distribution, and many others. You are free to select your domains, but if there are already **6 groups** working on a single domain, we would ask you to select another. To ensure this, please follow the naming convention - Domain (X), where “Domain” is your selected domain and “X” is your selection order. For example, the first team to select Health should write Health (1). The second team that decides to have a use case in the health domain should write Health (2). Please check the Excel sheet to make sure your domain can be selected. No domain should have Domain (7).

You can find real AI use cases on this website. You can also find these use cases in news articles. Let’s say you have chosen a domain, searching for news articles on “Domain AI use” or “Domain AI use harms” can jumpstart your search. You can also search Google Scholar for recent research on these topics to find possible AI use cases. You can use AI tools to brainstorm possible use cases but make sure they are real-life cases.

The deadline for selecting a domain is 24th March, 23:55.

B. Review the AI Use Case

After you select your domain and specific topic, research potential problems and how using AI systems in these situations can cause harm. Collect research papers and news articles around your topic to review the situation. While reviewing, you could start by answering the following questions:

- In what ways can using AI for this task be beneficial/problematic?
- What kinds of biases might there be? Representational, historical, …
- What kinds of problems can arise regarding the data collection or other data-related concerns?
- Who is harmed by the system? Identify the stakeholders in the situation and list how they are impacted by the system.
- Any issues with privacy?
- Any fairness issues?
- Any other problems?
- How prominent/widespread is the use of AI in this task?
- Is there any human oversight of the task?
- Find real-life examples to support your review.

Your review should not only be confined to answering these questions. Bring your own lens and discuss how this would be problematic to individuals, society, countries, the world, etc.

C. Ethical System Design

After the topic selection and review, we want you to design a better system to reduce harm. What would make the AI use in your selected topic more explainable? How would you design accountability into the process?

Each member will choose one of the AI ethics principles lectured in the course or from the following resources. You can also get an idea of how to design systems with these principles from these documents.

https://www.unesco.org/en/artificial-intelligence/recommendation-ethics
https://www.industry.gov.au/publications/australias-artificial-intelligence-ethics-principles/australias-ai-ethics-principles
https://www.ibm.com/think/topics/ai-ethics

After choosing your individual principle, design a system that would comply with the principle you have chosen. Think about the following questions while designing:

- What would the system look like? Show diagrams of workflows. (logic flowcharts, UI/UX mockups, technical specifications such as pseudo-code/algorithmic descriptions)
- Why would implementing that principle help?
- Who would benefit from implementing this principle in your selected topic?
- What are the challenges of implementing such a system? Investigate economic and structural costs and constraints.
- What happens when your selected principle design clashes with another ethical principle (e.g., transparency could lower the privacy protection)? Please do a trade-off analysis.
- Support your design with relevant literature. How did you come up with the design? Give proper citations (Suitable sources for the literature are given below).
- Even if you had an original design idea, find out whether it has been tried before and how well it works.

We expect each member to choose a different principle. Make sure you coordinate in selecting your principles.

All groups* will select an *extra* principle as a group and design that principle into the system together.

*4-person PG teams will implement all of their chosen principles, so they do not need to select an extra one.

D. System Implementation

Create a working prototype of the design you have suggested as a group. You can find a working code from somewhere else (paperswithcode/huggingface, GitHub) and implement the selected principle design on it. You can create a small proof-of-concept prototype of the AI use case you have selected and add the principle design on top of that. You can use toolkits like the IBM AI Explainability toolkit or third-party libraries to implement your system. You are free to use other people's code (if their license permits) as long as you disclose what is written by you vs others.
4-person PG teams will implement this for all four principles they have selected.

E. Suitable Literature Sources

You will need related literature (research papers) to identify your use case, review your use case, and design the system with ethical principles in mind. You are free to select your papers from any venue. However, papers from the following venues undergo a more rigorous review process, which can result in higher-quality papers.

1. Papers from Q1/Q2 journals in the Scimago, including:
    a. Mind and Machines
    b. AI and Society
    c. Big Data and Society
    d. ACM Transactions on …. type journals
    e. IEEE Transactions on … type journals
2. Recognized conferences in the field, such as
    a. FAccT,
    b. AIES,
    c. Or other rank A/B conferences with ethics-related tracks such as NeurIPS, CVPR, ACL, … (Select from A/B ranks in the conference ranks here.)

Go take a look at some of the venues and explore. It will help you find AI use cases and review them.

F. Deliverables

- For any type of submissions, please write your group number in the submitted files.
  Ex: GroupXX_final_presentation, GroupXX_final_report, GroupXX_final_code.
- Milestone and final presentation slots will be opened in the Excel sheet, and you will be able to select your slot with your assigned evaluator. Assigned evaluators will be decided after the group formations and domain selections are complete.
- You will make a presentation in your slot, but you must submit your slides along with your report and code by the deadline specified in the project submission (e.g., submit .pptx files too).
- The presentations should summarize the reports. We should be able to understand the main points of your work from the presentation. The presentation should follow the report structure.
- Every team member must come to the presentation, even if they will not be the presenter.
- You should cite existing work in your slides and reports (no hallucinations).
- Please submit your code for the final deliverable in SUCourse. Please submit a clearly commented Jupyter Notebook with output logs. Do not clear the output logs!

Milestone Deliverables (8th April)

For the milestone, select your domain and topic, and give the review you have written as detailed in Sections A and B. You will also write which specific AI ethics principles each member chooses to design for the remainder of the project, as well as your selected principle as a group. You will write your report as detailed below in Section G. Please keep to the page limit. You will also make a **5-minute presentation** to your assigned TA.

1. Group Report
2. Group Presentation - submit your slides in the assignment and present in your time slot

Final Deliverables (18th May)

For the final, you will submit a **group report** summarizing the selected use case and your review, as well as the final system design with all selected principles. You will also share your implemented principle as a group. There will also be **individual reports** in which each group member will write their design process for the selected AI ethics principle. There will be **group presentations** or **video submissions**. The final presentations will be **10 minutes** sharp. We will stop you if you go past this time. If you submit a video, you do not need to make a separate presentation, but videos should also comply with the presentation time limits. In the presentation, make sure to follow the guidelines, include a summary of the proposed designs for each selected principle, discuss the integration of principles together, and show the implementation of the group principle.
Feel free to be creative with your video submission!

1. Group Report
2. Group Presentation (submit slides) or Video Submission
3. Individual Report
4. Project Code - Jupyter Notebook

G. Report Formats

Please write the milestone and project reports in a clear, academic format. The following section contents are for guiding purposes. They are necessary but might not be sufficient. You should include the requested information in your report as a minimum, and you should add any other relevant information you deem necessary.

Milestone Report

Your milestone report should have an Introduction, a Review of the AI Use Case, Next Steps (summary on who selected what AI ethics principle and why that particular principle, as well as the group principle), Conclusion. The Review of the AI use section should be written as detailed in **Section B**, answering the questions posed there.

Milestone reports should be no more than **4 pages** (excluding references and appendices).

Final Report

In the project report, you are asked to have the following sections:

1. Introduction

You should give your selected domain and topic. This is an introduction to your chosen AI use case, the review, and the most important parts of your suggested design. The introduction should give a quick overview of your project.

2. Review of The AI Use Case

This could be a summary of your milestone report. What AI use case are you using? Which papers did you get help from while reviewing the AI use case? Explain why you use the selected AI use case and papers. Give the summary of the review you have done as detailed in **Section B**.

3. Ethical System Design

State the AI Ethics principles selected by each member of the team. For each principle, give a summary of your proposed design. Why have you selected this particular design, and what papers have you used while designing, … Basically, this should be a quick summary of each member's individual report. How will the integration of different principles work in the overall design? Can they be integrated?
State your selected group principle here. Write the process for this common principle in more detail, more like the structure of the individual reports.

4. System Implementation

Give details on your implementation. What technologies have you used to implement your system? Was it implemented on an already existing system, or have you implemented a proof-of-concept code to add-on the principle? Have you used any third-party libraries or code? What were the challenges you faced while implementing the proposed system? How well does it work? Evaluate your system. Discuss the limitations of your implementation.

5. Discussion

Please discuss these points in this section:

- The selection of your AI use case and how that affected the design process
- What are the advantages/disadvantages of your overall design?
- Comparison of your overall design to the existing systems in prior literature.
- Limitations of the proposed system.
- Possible improvements on the overall design if you had more time and resources.

6. Conclusion

Give another quick overview of the project and list the most important design/discussion points to conclude the report.

—-------

The reports should be no more than **8 pages** (excluding references and appendices). Make sure to include the most important points in the main report and leave the extras in the appendix.

Individual Reports

In the individual reports, you should have the following sections: Introduction, Ethical System Design (Principle Name), Implementation (if you are in a 4-person PG group), Discussion, and Other Remarks.

Please detail your selection of AI ethics principle, design process, your final design with components, and discussion of your design in terms of advantages/disadvantages. Ethical System Design (Principle Name) should be written as detailed in **Section C**, answering the questions posed there.

Please state your unique contribution to the group's principle implementation. Your contribution might not necessarily be coding-based, as there are people from different programs. You might have contributed to the literature review for the implementation. That is also a contribution. State your background if it is not CS/DSA. Individual reports should be no more than **4 pages** (excluding references and appendix).

H. Grading

- 30% from the milestone presentation and report
- 20% from your final presentation and group report
- 50% from your individual report + your contribution

I. AI Use

You are allowed to use AI systems to spell-check your English and improve the flow of your report.

You are **NOT** allowed to write your report from scratch using an AI system. Write the report first, and then improve on it with AI on your wording and flow.

You are allowed to use AI systems to get help with your code. Give details on which parts were written with an AI tool, which parts were taken from another person's code, and what your contribution is.